SHIVAM SRIVASTAVA

12814 University Club Dr, Tampa, Florida 33612
8133379597 ⋄ shivam4@usf.edu ⋄ LinkedIn

EDUCATION

University of South Florida — Tampa, FL — Aug. 2019 - May 2021
Master of Science in Computer Science
GPA: 3.95

JSS Academy of Technical Education — Noida, India — Aug. 2012 - June 2016
Bachelor of Technology in Computer Science & Engineering
GPA: 3.75

PUBLICATION

Recognizing Emotion in the Wild using Multimodal Data
***S. Srivastava**, S. Aathreya, S. Hinduja, Sk R. Jannat, H. Elhamdadi, and S. Canavan.*
International Conference on Multimodal Interaction, 2020 [LINK]

PROGRAMMING SKILLS

Language	Python, Java, Javascript
Databases	MySQL, PostgreSQL, MongoDB, Redshift(AWS)
AWS Services	Lambda, Kinesis, SQS, SNS, S3, EC2, Athena, EMR, Glue
Web Development	Django, React JS, HTML/CSS
Big Data	Hadoop, Spark, Kafka, Airflow
Deep Learning	TensorFlow, PyTorch, Numpy
Deployment	Nginx, Gunicorn, Ansible

WORK EXPERIENCE

University of South Florida — Jan. 2020 - Present
Teaching Assistant and Student Researcher

- Teaching Assistant for Introduction to Theory of Algorithms (COT 6405) instructed by Dr. Shaun Canavan.
- Ongoing Research: Automatic emotion and facial expression detection using non verbal cues.

Delhivery — June 2017 – June 2019
Software Engineer

- Responsible for building a scalable enterprise level **data warehouse** enabling teams to get historical as well real-time state of all the packages and courier vehicles across 17,500+ centers in India.
- Managed a multi-node **Redshift** cluster, handling 2000+ queries serving different dashboards daily. Separating OLAP and OLTP queries **reduced production downtime by 2X** and also **improved the average time for analytical queries by 30%**. Collaborated with the Infrastructure team frequently for ensuring scalability, security and availability of the cluster and other resources.
- Created an **ETL pipeline** using **AWS Big Data Ecosystem** and **Apache Airflow** that validates, clean and store the data. Implemented an **Apache Spark** job that converts data on S3 in row storage JSON format to **columnar storage** Parquet format which **reduced the storage cost by 80% on S3 and approximately 70% data scans on queries**, reducing Athena's (AWS's query service) per-TB-scanned billing cost.
- Implemented **unit testing** using Pytest and an offline **integration testing pipeline** using localstack/moto library which removed the dependency of creating and maintaining resources for test environment, thus **cutting total resource's cost by 25%**. Concurrently, assisted in designing architecture and schema for distributing, sorting and storing data such that it tunes query performance.

Neuron June 2016 – May 2017

Software Engineer

- Created and managed Neuron's developer platform which offered multiple natural language processing based services to users according to the subscription opted. Responsible for designing and developing server side **RESTful services** using **Django REST Framework**.
- Created a scalable and interactive **Chatbot** and it's admin panel **in 2 months** to build and store surveys that can be shared easily on multiple social media platforms and sites, replacing the usual form type questionnaire format for clients. Used **Django, React JS** as the web frameworks and **MySQL, MongoDB** as the databases.
- Followed **agile** methodology with automated unit testing and **CI/CD** using Jenkins to ensure a seamless deployment of code on production and **Git/BitBucket** as the version control tool.
- Worked closely with clients, AI team and Senior Developers for requirement gathering, planning and implementation of the projects with focus on **high availability, low latency and scalability**.

INTERNSHIPS

Neuron Dec. 2015 - Apr. 2016

- Integrated Hindi news Ad recommendation system with clients applications to fetch and display other related news as well along with the main news.
- Integrated a JavaScript library, fingerprint.js, on client's machine for anonymous browser detection and hence throttling user based on unique fingerprint/ID.

Algoscale May 2015 - July 2015

- Devised a web based analytics tool to fetch and analyze the keywords related to a YouTube channel.
- Developed an admin portal for clients using Django's MVT architecture using which clients can get the result in their desired text format.

FratMart Sep. 2014 - Mar. 2015

- Developed an online peer-to-peer marketplace for buying and selling used stuff (books, study material, novels, electronics etc), finding flats/flatmates or arrange carpools to college.
- Responsible for the development of the whole product from scratch, managing the server, database management and other both backend and frontend related tasks.

SELECTED PROJECTS

- **JSS Infoconnect:** Revamped the old PHP-based web architecture of Student-Faculty interaction platform using modern framework and libraries Django and Angular, serving 10,000+ Daily Active Users
- **Food Genie:** Android based app for finding ratings of food items per restaurant. Used Natural Language Processing on user and critic reviews of restaurants to analyze the frequency and sentiment of users towards a particular item and calculate the rating.
- **PyURL Shortener:** Create short and custom URLs that can be easily shared.

SELECTED COURSEWORK

Digital Image Processing, Computer Vision, Pattern Recognition, Introduction to AI, Introduction to Robotics, Distributed Systems, Design and Analysis of Algorithms, Theory of Computation, Software Engineering

ACHIEVEMENTS AND EXTRA CURRICULAR

- **3rd place,** Eighth Emotion Recognition in the Wild Challenge (EmotiW 2020).
- **Winner,** Web Developemt @ Zealicon Tech fest, JSS Noida.
- **Runner-up,** GDG-Lingaya's App Development Hackathon, Delhi.
- **Organiser,** Google Developer Group, JSS Noida.
- **Organiser,** Hackathon 2015, JSS Noida.
- **Web Head,** Microsoft Mobile Innovation Lab, JSS Noida.